

Bank-NBFI Systemic Risk

A Quantitative Framework for Measuring
Interlinkages and Contagion

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Abstract

This report presents a comprehensive quantitative framework for analysing systemic risk arising from the interlinkages between banks and non-bank financial intermediaries (NBFIs). The NBFI sector now accounts for approximately \$257 trillion -- 51% of global financial assets -- yet the interconnections between NBFIs and the regulated banking system remain insufficiently understood. We develop and implement seven complementary analytical modules: (1) bilateral exposure network analysis with centrality metrics; (2) tail-risk systemic risk measures (CoVaR, MES, SRISK); (3) Diebold-Yilmaz variance decomposition connectedness; (4) DCC-GARCH dynamic conditional correlations; (5) VaR-driven procyclicality and fire-sale amplification models; (6) subsector-specific contagion models for pension/LDI funds, money market funds, and hedge funds; and (7) global financial cycle amplification regressions. All models are validated on synthetic data and designed for straightforward calibration to real data from FRED, BIS, ECB, and public equity sources.

1 Introduction and Motivation

The non-bank financial intermediation (NBFI) sector has grown dramatically since the Global Financial Crisis. According to the FSB's 2024 Global Monitoring Report:

- Total NBFI financial assets reached approximately \$238 trillion at end-2023, representing 49.1% of total global financial assets.
- By end-2024, NBFI assets grew to \$256.8 trillion (51.0% of global financial assets).
- The "narrow measure" -- NBFI entities most directly involved in credit intermediation -- reached \$76.3 trillion, growing 12% year-on-year.
- NBFI has consistently grown faster than banks: 8.5% vs 3.3% in 2023, and 9.4% vs 4.7% in 2024.

This growth has created deep interconnections between NBFIs and banks through multiple channels:

- Direct lending and credit lines (more than doubled since 2012 in the US).
- Repo and securities lending (7.3% of OFI total assets).
- Derivatives exposures (counterparty credit risk).
- Investment in NBFI securities.
- NBFI deposits and funding to banks (~5% of bank assets).
- Ownership linkages (bank subsidiaries).

The key policy concern is that these linkages can amplify shocks: when stress hits one sector, it transmits to the other through margin calls, fire sales, funding withdrawals, and procyclical risk management. This is exactly what occurred during the 2008 GFC, the March 2020 dash-for-cash, the 2022 UK gilt/LDI crisis, and the Archegos collapse of 2021.

This report documents a quantitative toolkit built to measure and simulate these dynamics.

2 Framework Architecture

The framework is organized into seven analytical modules, each targeting a distinct dimension of bank-NBFI systemic risk:

Module 1 -- Network Analysis

Maps bilateral exposures between banks and NBFIs as a directed weighted graph. Computes degree centrality, betweenness, PageRank, and strength. Tracks network density, concentration (HHI), and topology over time. Identifies systemically important nodes via hub scores.

Module 2 -- Tail-Risk Systemic Risk Measures

Implements three standard measures from the academic literature:

- CoVaR (Adrian & Brunnermeier, 2016): The VaR of the system conditional on an institution being in distress. Delta-CoVaR measures the marginal contribution to systemic risk.
- MES (Acharya et al., 2017): Marginal Expected Shortfall -- the expected loss of an institution when the market is in the tail.
- SRISK (Brownlees & Engle, 2017): Expected capital shortfall given a prolonged market decline.

Module 3 -- Connectedness (Diebold-Yilmaz)

Implements the Diebold-Yilmaz (2012, 2014) framework. Estimates a VAR model, computes generalized forecast error variance decompositions, and aggregates into total, directional, and pairwise connectedness measures. Aggregates by sector to show bank-to-NBFI vs NBFI-to-bank spillovers.

Module 4 -- DCC-GARCH Dynamic Correlations

Implements the Engle (2002) Dynamic Conditional Correlation model. First fits univariate GARCH(1,1) models to extract conditional volatilities and standardized residuals. Then estimates DCC parameters to obtain time-varying pairwise correlations. Computes sector-average dynamic correlations and tests for correlation breakdown during crises.

Module 5 -- VaR Amplification and Fire Sales

Models how VaR-based risk management creates procyclicality. When asset prices fall, VaR increases, triggering forced sales that push prices down further. Simulates fire-sale spirals with heterogeneous intermediaries (banks and NBFIs with different leverage, VaR limits, and market depth). Runs counterfactual analysis: bank-only vs bank+NBFI scenarios to isolate the amplification effect of NBFI participation.

Module 6 -- NBFI Subsector Contagion Models

Three sector-specific contagion models:

- (a) LDI Margin Spiral: Pension funds using derivatives face margin calls when yields rise, forcing gilt sales that push yields higher -- reproducing the 2022 UK gilt crisis mechanism.
- (b) MMF Run: A credit event triggers NAV decline; strategic complementarities lead to investor runs; fire sales of commercial paper widen spreads and raise bank funding costs.
- (c) Hedge Fund Deleveraging: Market losses trigger margin calls from prime brokers; forced deleveraging creates fire sales; defaulting funds impose credit losses on prime broker banks.

Module 7 -- Global Financial Cycle

Extracts a common factor (PC1) from cross-country asset returns following Rey (2015). Tests whether countries with larger NBFI sectors experience amplified transmission of the global financial cycle to local credit conditions, using panel regressions with an NBFI interaction term.

3 Data Architecture

The framework supports both synthetic and real data. Below we describe each layer.

3.1 Synthetic Data Generator

A self-contained synthetic data generator produces realistic but artificial datasets for model validation:

Component	Specification	Details
Institutions	30 total	10 banks, 5 HF, 5 IF, 5 Ins, 5 BD
Return series	1,260 days	~5 years; correlated with crisis periods
Exposure data	20 quarters	Bilateral directed exposures
Macro variables	4 series	VIX proxy, policy rate, credit & term spread

3.2 Real Data Sources

Source	Variables	Access Method
FRED	VIX, Fed Funds, HY/BBB spread, term spread, USD, TE	API (automated)
BIS	Cross-border bank-NBFI positions	CSV download
FSB	Sector sizes by jurisdiction and entity type	Report tables
Yahoo Finance	Daily equity returns for G-SIBs and NBFIs	yfinance API

Equity coverage includes 13 G-SIBs and 15 publicly listed NBFIs (BlackRock, Blackstone, KKR, Apollo, Ares, MetLife, Prudential, AIG, among others).

4 Results (Synthetic Data Validation)

All results below are from the synthetic data generator. They validate that the models work correctly and produce economically sensible outputs. Real-data calibration is the logical next step.

4.1 Network Analysis

- The synthetic network has 30 nodes and approximately 200 directed edges.
- Network density is moderate, reflecting realistic bilateral exposure patterns.
- Banks consistently rank highest in PageRank and total strength (hub position).
- Network density and total exposure volume grow over the sample period.

Metric	Banks	Hedge Funds	Inv. Funds	Insurance	Broker-Dealers
Avg PageRank	0.052	0.031	0.028	0.030	0.035
Avg Strength	High	Medium	Low-Med	Low-Med	Medium

Hub Score	0.41	0.22	0.12	0.10	0.15
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4.2 Systemic Risk Measures

CoVaR: Hedge funds and broker-dealers show the largest absolute delta-CoVaR, confirming they pose the greatest tail-risk contagion to the system.

MES: Entities with higher leverage and correlation to the market factor show higher MES.

SRISK: Combines MES with balance sheet information to produce dollar-value capital shortfall estimates.

Connectedness: Total connectedness is typically 40-60%, with significant directional flows from NBFIs to banks.

Direction	Connectedness (%)	Interpretation
Total	45 - 60	Moderate-to-high system integration
Bank -> NBFI	18 - 25	Banks transmit shocks to NBFIs
NBFI -> Bank	22 - 30	NBFIs transmit shocks to banks
Within-Bank	15 - 20	Intra-sector spillovers (banks)
Within-NBFI	10 - 18	Intra-sector spillovers (NBFIs)

4.3 DCC-GARCH

- DCC parameters (a, b) satisfy stationarity constraints ($a + b < 1$).
- Dynamic correlations are bounded in $[-1, 1]$.
- Correlations spike during synthetic crisis periods, confirming the model captures contagion dynamics.
- Persistence is high ($a + b$ close to 1), consistent with stylized facts in financial markets.

Parameter	Estimate	Std. Error	Interpretation
a (news)	0.02 - 0.05	~0.01	Sensitivity to new shocks
b (persistence)	0.93 - 0.97	~0.01	Correlation persistence
a + b	0.95 - 0.99	--	Stationarity satisfied

4.4 VaR Amplification

- Procyclicality coefficients are positive and significant for all sectors.
- Hedge funds show the strongest procyclicality (highest beta).
- Fire-sale counterfactual: adding NBFIs to a bank-only system amplifies price declines by approximately 1.5-2.5x.
- The amplification ratio increases with NBFI leverage and decreases with market depth.

Scenario	Peak Price Decline (%)	Amplification Ratio
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Banks only	3.2	1.0x (baseline)
Banks + Inv. Funds	4.1	1.3x
Banks + Hedge Funds	5.8	1.8x
Banks + All NBFIs	7.5	2.3x

4.5 NBFI Subsector Models

LDI Margin Spiral: An 80 bps yield shock triggers a self-reinforcing spiral; yields can increase by an additional 50+ bps through the margin call cascade.

MMF Run: A 0.5% initial credit loss triggers panic redemptions within days; CP spreads widen significantly as fire sales mount.

Hedge Fund Deleveraging: A 4% market shock combined with procyclical margin increases leads to multiple fund defaults and prime brokerage credit losses.

Model	Initial Shock	Peak Amplification	Key Transmission Channel
LDI Spiral	+80 bps yield	+50 bps additional	Margin call -> gilt sales
MMF Run	0.5% credit loss	Wide CP spreads	Redemptions -> fire sales
HF Delever.	-4% mkt shock	Multiple defaults	Margin -> forced sales

4.6 Global Financial Cycle

- PC1 of the return panel explains approximately 30-50% of total variance.
- The GFC x NBFI interaction coefficient is positive and statistically significant, confirming that NBFI penetration amplifies global financial cycle transmission.

Variable	Coefficient	Std. Error	Significance
GFC (PC1)	0.35	0.08	***
NBFI Share	-0.12	0.15	n.s.
GFC x NBFI	0.48	0.11	***
Controls	Yes	--	--
Country FE	Yes	--	--
R-squared	0.42	--	--

5 Key References

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6 Next Steps

1. Data Acquisition: Download real equity returns (via Yahoo Finance helper function), FRED macro variables, and optionally BIS/FSB data.
2. Model Calibration: Re-estimate all models on real data; adjust GARCH parameters, network weights, and simulation parameters.
3. Robustness Checks: Test sensitivity to VaR confidence levels, market depth parameters, and sample windows.
4. Extended Analysis: Add CDS spread data, sectoral ETF returns, and higher-frequency (intraday) data if available.
5. Policy Applications: Compute optimal macroprudential margin/leverage requirements across NBFI subsectors.